

14-subgroup: 子群 (Subgroup)

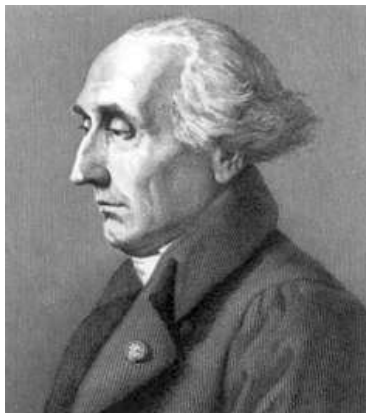
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Lagrange's Theorem



Joseph-Louis Lagrange (1736 ~ 1813)

Fundamental Homomorphism Theorem



Emmy Noether (1882 ~ 1935)

Lagrange's Theorem

Help us understand the structure of a group
via its subgroups/normal subgroups

Fundamental Homomorphism Theorem

Definition (Coset (陪集))

Suppose that $H \leq G$. For $a \in G$,

$$aH = \{ah \mid h \in H\}, \quad Ha = \{ha \mid h \in H\},$$

is called the **left coset** (左陪集) and **right coset** of H in G , respectively.

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2)\} \leq S_3$$

$$(1)H = H = (1\ 2)H \leq S_3$$

$$(1\ 3)H = \{(1\ 3), (1\ 2\ 3)\} = (1\ 2\ 3)H$$

$$(2\ 3)H = \{(2\ 3), (1\ 3\ 2)\} = (1\ 3\ 2)H$$

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$A_3 = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \leq S_3$$

$$(1)A_3 = (1\ 2\ 3)A_3 = (1\ 3\ 2)A_3 = A_3 \leq S_3$$

$$(1\ 2)A_3 = (1\ 3)A_3 = (2\ 3)A_3 = \{(1\ 2), (1\ 3), (2\ 3)\}$$

Theorem

Suppose that $H \leq G$, $a, b \in G$.

(1)

$$a \in aH$$

(2)

$$aH = H \iff a \in H$$

(3)

$$aH \leq G \iff a \in H$$

(4)

$$aH = bH \iff a^{-1}b \in H$$

(5)

$$\forall a, b \in G \ (aH = bH) \vee (aH \cap bH = \emptyset)$$

(6)

$$|aH| = |H| = |bH|$$

$$aH = H \iff a \in H$$

$$aH = H \implies a \in H$$

$$a \in H \implies aH = H$$

$$a \in aH = H$$

$$aH \subseteq HH \triangleq \{hh' \mid h, h' \in H\} = H$$

$$H = (aa^{-1})H = a(a^{-1}H) \subseteq aH$$

$$aH \leq G \iff a \in H$$

$$a \in H \implies aH \leq G$$

$$aH \leq G \implies a \in H$$

$$a \in H \implies aH = H \leq G$$

$$a \in aH \implies a^{-1} \in aH \implies e \in aH$$

$$\exists h \in H \ e = ah \implies a = h^{-1} \in H$$

$$aH = bH \iff a^{-1}b \in H$$

$$a^{-1}b \in H \iff a^{-1}bH = H$$

$$aH = bH \implies a^{-1}aH = a^{-1}bH \implies a^{-1}bH = H$$

$$a^{-1}bH = H \implies a(a^{-1}bH) = aH \implies bH = aH$$

$$\boxed{\forall a, b \in G \ (aH = bH) \vee (aH \cap bH = \emptyset)}$$

$$\forall a, b \in G \ (aH \cap bH \neq \emptyset \rightarrow aH = bH)$$

Take any $g \in aH \cap bH$.

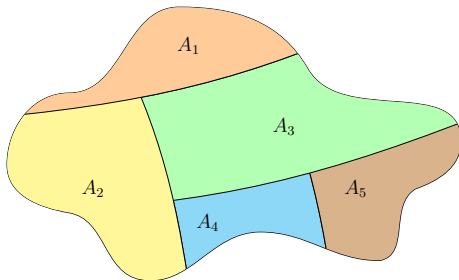
$$\exists h_1, h_2 \in H \ (ah_1 = g = bh_2)$$

$$aH = a(h_1H) = (ah_1)H = (bh_2)H = b(h_2H) = bH$$

$$|aH| = |H| = |bH|$$

$$\sigma : aH \rightarrow H, \quad ah \mapsto h$$

A balanced partition of G by its subgraph H



$$\forall a, b \in G \ (aH = bH) \vee (aH \cap bH = \emptyset)$$

$$|aH| = |H| = |bH|$$

Definition (Index (H 在 G 中的指标/指数))

$$G/H = \{gH \mid g \in G\}$$

$$[G : H] \triangleq |G/H|$$

Theorem (Lagrange's Theorem)

Suppose that $H \leq G$. Then

$$|G| = [G : H] \cdot |H|$$

Theorem

$$H \leq G \implies |H| \mid |G|$$

There are *no* subgroups of order 5, 7, or 8 of a group of order 12.

Theorem

Let G be a finite group, $a \in G$. Then the order of a divides the order of G .

Theorem

- ▶ *There are only 2 groups of order 4.*
- ▶ *There are only 2 groups of order 6.*

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2)\} \leq S_3$$

$$(1)H = H = (1\ 2)H$$

$$(1\ 3)H = \{(1\ 3), (1\ 2\ 3)\} = (1\ 2\ 3)H$$

$$(2\ 3)H = \{(2\ 3), (1\ 3\ 2)\} = (1\ 3\ 2)H$$

$$H(1) = H = H(1\ 2)$$

$$H(1\ 3) = \{(1\ 3), (1\ 3\ 2)\} = (1\ 3\ 2)H$$

$$H(2\ 3) = \{(2\ 3), (1\ 2\ 3)\} = (1\ 2\ 3)H$$

It is possible that $aH \neq Ha$.

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \leq S_3$$

$$(1)H = (1\ 2\ 3)H = (1\ 3\ 2)H = H \leq S_3$$

$$(1\ 2)H = (1\ 3)H = (2\ 3)H = \{(1\ 2), (1\ 3), (2\ 3)\}$$

$$\forall a \in S_3 \ aH = Ha$$

Definition (Normal Subgroup (正规子群))

Suppose that $H \leq G$. If

$$\forall a \in G \ aH = Ha,$$

then H is a **normal subgroup** of G , denoted $H \triangleleft G$.

$$aH = Ha \not\Rightarrow \forall h \in H \ ah = ha$$

$$aH = Ha \Rightarrow \forall h \in H \ \exists h' \in H. \ ah = h'a$$

Theorem

$$H \triangleleft G \iff \forall a \in G, h \in H \ aha^{-1} \in H$$

$$\begin{aligned} aH = Ha &\implies aHa^{-1} = (Ha)a^{-1} = H(aa^{-1}) = H \\ &\implies aHa^{-1} \subseteq H \\ &\implies \forall h \in H \ aha^{-1} \in H \end{aligned}$$

$$aha^{-1} \in H \implies ah = (aha^{-1})a \in Ha \implies aH \subseteq Ha$$

$$a^{-1}ha = a^{-1}h(a^{-1})^{-1} \in H \implies ha \in aH \implies Ha \subseteq aH$$

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \triangleleft S_3$$

$$\forall \sigma \in S_3, \tau \in H. \sigma\tau\sigma^{-1} \in H$$

Theorem

$$\sigma\tau\sigma^{-1} = \begin{pmatrix} \sigma(1) & \sigma(2) & \dots & \sigma(n) \\ \sigma(\tau(1)) & \sigma(\tau(2)) & \dots & \sigma(\tau(n)) \end{pmatrix}$$

$$(1\ 2)(1\ 2\ 3)(1\ 2)^{-1} = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 3 & 2 \end{pmatrix} = (1\ 3\ 2)$$

Definition (正规子群的陪集)

Suppose that $H \triangleleft G$.

$$G/H = \{aH \mid a \in G\}$$

is the **set of cosets** of H in G .

Definition (Quotient Group (商群))

Suppose that $H \triangleleft G$. Define

$$aH \cdot bH = (ab)H.$$

Then $(G/H, \cdot)$ is a group, called the **quotient group** of G by H (denoted G/H).

$aH \cdot bH = (ab)H$ is well-defined

$$aH = a'H \wedge bH = b'H \implies aH \cdot bH = a'H \cdot b'H$$

结果与代表元的选取无关

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \triangleleft S_3$$

$$G/H = \{(1)H, (1\ 2)H\}$$

$$G = \mathbb{Z} \quad H = 6\mathbb{Z} \triangleleft G$$

$$G/H = \{0 + H, 1 + H, \dots, 5 + H\}$$

Definition (Homomorphism (同态))

Let $(G, \cdot)$ and $(G', *)$ be two groups. Let ϕ be a function such that

$$\forall a, b \in G. \phi(ab) = \phi(a)\phi(b).$$

Then ϕ is a **homomorphism** from G to G' .

If ϕ is a bijection, then G and G' are called **isomorphic**.

$$\phi : G \cong G'$$

$$\phi : \mathbb{Z} \rightarrow \mathbb{R}^*$$

$$n \mapsto (-1)^n$$

$$\phi(m+n) = (-1)^{m+n} = \phi(m)\phi(n)$$

$$\phi : \mathbb{Z} \rightarrow \mathbb{Z}_6$$

$$a \mapsto [a]_6$$

$$\phi(a + b) = [a + b]_6 = \phi(a) + \phi(b)$$

$$\begin{aligned}\phi : \mathbb{R}[x] &\rightarrow \mathbb{R}[x] \\ f(x) &\mapsto f'(x)\end{aligned}$$

$\mathbb{R}[x]$: 全体实系数多项式关于多项式的加法构成的群

$$\phi(f(x) + g(x)) = (f(x) + g(x))' = \phi(f(x)) + \phi(g(x))$$

Theorem

Suppose that ϕ is a homomorphism from G to G' .

Let e and e' be identities of G and G' , respectively.

$$(1) \quad \phi(e) = e'$$

$$(2) \quad \phi(a^{-1}) = (\phi(a))^{-1}$$

$$e'\phi(e) = \phi(e) = \phi(ee) = \phi(e)\phi(e) \implies \phi(e) = e'$$

$$\phi(a)\phi(a^{-1}) = \phi(aa^{-1}) = \phi(e) = e' = \phi(a)(\phi(a))^{-1}$$

Theorem

Suppose that ϕ is a homomorphism from G to G' .

(1)

$$H \leq G \implies \phi(H) \leq G'$$

(2)

$$H \triangleleft G \implies \phi(H) \triangleleft \phi(G)$$

(3)

$$K \leq G' \implies \phi^{-1}(K) \leq G$$

(4)

$$K \triangleleft G' \implies \phi^{-1}(K) \triangleleft G$$

Definition (核 (Kernel))

Suppose that ϕ is a homomorphism from G to G' .

Let e' be the identity of G' .

$$\phi^{-1}(\{e'\}) = \{a \in G \mid \phi(a) = e'\}$$

is the **kernel** of ϕ , denoted $\text{Ker } \phi$.

$$\text{Ker } \phi \triangleleft G$$

$$\phi : \mathbb{Z} \rightarrow \mathbb{R}^*$$

$$n \mapsto (-1)^n$$

$$\text{Ker } \phi = 2\mathbb{Z}$$

$$\phi : \mathbb{Z} \rightarrow \mathbb{Z}_6$$

$$a \mapsto [a]_6$$

$$\text{Ker } \phi = 6\mathbb{Z}$$

$$\begin{aligned}\phi : \mathbb{R}[x] &\rightarrow \mathbb{R}[x] \\ f(x) &\mapsto f'(x)\end{aligned}$$

$\mathbb{R}[x]$: 全体实系数多项式关于多项式的加法构成的群

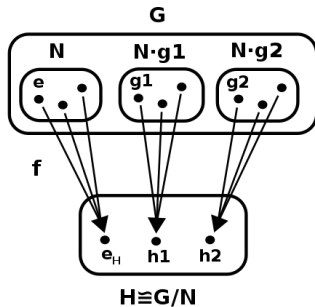
$$\text{Ker } \phi = \mathbb{R}$$

Theorem (Fundamental Homomorphism Theorem)

Suppose that ϕ is a homomorphism from G to G' . Then

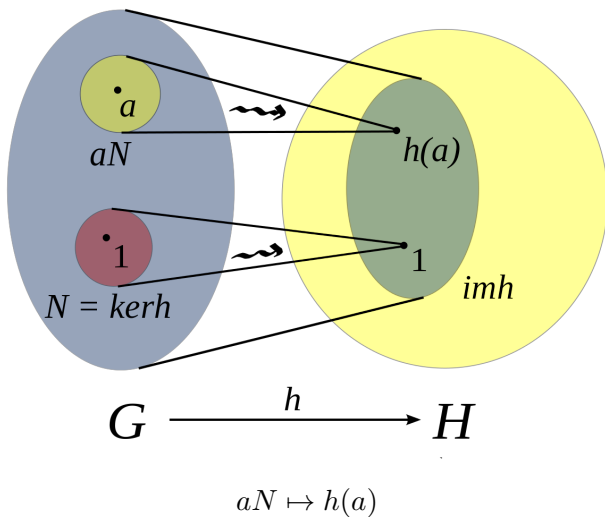
$$G/\text{Ker } \phi \cong \phi(G).$$

同态核可以看作群 G 与其同态像 $\phi(G)$ 之间相似程度的一种度量



$$N = \text{Ker } \phi$$

$$G/(N \triangleq \text{Ker } h) \cong (h(G) \triangleq \text{im } h)$$



$$\phi : \mathbb{Z} \rightarrow \mathbb{R}^*$$

$$n \mapsto (-1)^n$$

$$\text{Ker } \phi = 2\mathbb{Z}$$

$$\mathbb{Z}/(2\mathbb{Z}) = (2\mathbb{Z}, 2\mathbb{Z} + 1) \cong \phi(\mathbb{Z}) = (-1, 1)$$

$$\phi : \mathbb{Z} \rightarrow \mathbb{Z}_6$$

$$a \mapsto [a]_6$$

$$\text{Ker } \phi = 6\mathbb{Z}$$

$$\mathbb{Z}/(6\mathbb{Z}) = \{0 + H, 1 + H, \dots, 5 + H\} \cong \phi(\mathbb{Z}) = \mathbb{Z}_6$$

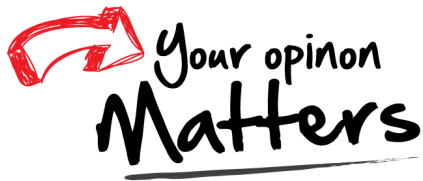
$$\begin{aligned}\phi : \mathbb{R}[x] &\rightarrow \mathbb{R}[x] \\ f(x) &\mapsto f'(x)\end{aligned}$$

$\mathbb{R}[x]$: 全体实系数多项式关于多项式的加法构成的群

$$\text{Ker } \phi = \mathbb{R}$$

$$\mathbb{R}[x]/\mathbb{R} \cong \phi(\mathbb{R}[x]) = \mathbb{R}[x]$$

Thank
You!



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